

Total no. of Pages:

Roll no.

Degree: M. Tech 2nd Semester
END-SEMESTER EXAMINATION, JUNE – JULY, 2026

Course Title: CMOS DIGITAL INTEGRATED CIRCUITS

Course Code: ECEVE07 ECECE07

Duration: 3 Hours

Max. Marks: 50

Note: - Attempt all questions in given order only. Missing data/information (if any), may be suitable assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q1.	Attempt any 2 parts of the following		
1.(a)	Explain the operation of a MOS transistor in the cutoff, linear and saturation regions. Derive the drain current equation for the saturation region.	5	CO1
1.(b)	Explain the effects of MOSFET scaling on device performance. Discuss any three small-geometry effects.	5	CO1
1.(c)	Derive the expression for dynamic power dissipation in CMOS circuits. Discuss the factors affecting propagation delay.	5	CO5
Q2.	Attempt any 2 parts of the following		
2.(a)	Design a two-input CMOS NOR gate using complementary CMOS logic and explain its operation.	5	CO2
2.(b)	Compare Static CMOS, Pseudo-NMOS and CMOS Transmission Gate Logic with respect to area, speed, power consumption and noise immunity.	5	CO5
2.(c)	Explain the Logical Effort Theory. Derive the logical effort of a CMOS inverter and discuss its significance in transistor sizing.	5	CO2
Q3.	Attempt any 2 parts of the following		
3.(a)	Explain the bistability principle and discuss the operation of static CMOS latches.	5	CO3
3.(b)	Draw and explain the operation of a Master-Slave D Flip-Flop.	5	CO3
3.(c)	Define setup time, hold time, clock skew and clock jitter. Explain their impact on synchronous digital systems.	5	CO3
Q4.	Attempt any 2 parts of the following		
4.(a)	Explain the working principle of Dynamic CMOS logic with suitable timing diagrams	5	CO3
4.(b)	Discuss the charge-sharing problem in dynamic CMOS circuits. Explain suitable techniques to overcome it.	5	CO3

4.(c)	Explain the following: • Ratioed and Ratioless Logic • Voltage Bootstrapping	5	CO3
Q5.	Attempt any 2 parts of the following		
5.(a)	Compare ROM, SRAM, and DRAM.	5	CO4
5.(b)	Explain the architecture and operation of an SRAM cell.	5	CO4
5.(c)	Explain the significance of noise margins in CMOS logic circuits.	5	CO1